

Decimal to binary, past the point

Name: _____ Goes with the September 30 lecture. Work alone or in pairs. Show your table, and check every answer by place value.

place is worth	32	16	8	4	2	1	.	1/2	1/4	1/8	1/16
as a decimal	32	16	8	4	2	1	.	0.5	0.25	0.125	0.0625

A. Whole numbers

1. Convert 26 to binary. Divide by 2 until the quotient is 0; read the remainders **last to first**.

2. $101101_2 = \rule{1cm}{0.4pt}$ (decimal)

Divide	Quotient	Remainder
$26 \div 2$		

$26 = \rule{1cm}{0.4pt}_2$

B. Past the point

1. $10.1_2 = \rule{1cm}{0.4pt}$ (decimal)

3. Convert 0.8125 to binary. Double, copy the whole-number bit, keep the fraction part; stop when Keep is 0. Read the bits **first to last**.

2. $0.0011_2 = \rule{1cm}{0.4pt}$ (decimal)

Double	Bit	Keep
$0.8125 \times 2 =$		

$0.8125 = \rule{1cm}{0.4pt}_2$

C. One that never ends

Run repeated doubling on 0.3. Stop when a Keep value repeats one you already wrote.

1. Which Keep value came back? Write 0.3 in binary using ...

Double	Bit	Keep
$0.3 \times 2 =$		

2. Why can't a computer store 0.3 exactly?